Whose r^*?

Belief Transmission, Intercept Disagreement, and the Price of Long Bonds

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1 The question

The federal funds rate is an overnight rate. The ten-year Treasury yield is not, and its thirty-year decline is usually attributed to forces outside monetary policy — demographics, productivity, the global supply of safe assets. Hillenbrand (2025) shows that this decline accumulated almost entirely inside three-day windows around FOMC announcements, and argues the Fed was revealing its own estimate of the long-run equilibrium rate. That interpretation makes a specific structural claim that has not been examined. Write the policy rule as

$$i_t \;=\; \underbrace{r^*_t + \pi^*_t}_{\text{intercept}} + \phi_\pi(\pi_t - \pi^*_t) + \phi_x x_t + v_t. \tag{1}$$

r^* is the *intercept* of the reaction function. So "the Fed reveals r^*" means the Fed is telling markets about the constant in its own rule, and disagreement between the Fed and the market about r^* is disagreement about that constant. Bauer, Pflueger & Sunderam (2024) estimate the market's perceived *slope* ϕ_π from forecaster panels and show it prices term premia; they absorb the intercept into forecaster fixed effects and say so explicitly. The intercept has been assumed known, assumed common, or differenced away.

> **This project asks: how does the Fed move beliefs about the intercept of its own reaction function, and what does disagreement about that intercept do to the price of long bonds?**

The question decomposes into three parts:

1. **Transmission.** Does the Fed's published longer-run projection move market participants' beliefs about the long-run rate, and in which direction does causality run?
2. **Pricing.** Does the revision in beliefs about the intercept pass through to long yields, and does it do so with the maturity signature of an endpoint shock rather than a stance shock?
3. **Disagreement.** Is uncertainty about the intercept — as opposed to its level — compensated in the term premium, and is the compensation asymmetric in the direction of the perceived error?

What the data have already removed

Three results reported below — Facts 2 and 3, and the measurement result in §8.4 — constrain this programme severely, and it is better to state the consequence at the front than to let a reader assemble it. Fact 3 shows that the yield response to be explained is concentrated *before* the published projection exists; in the SEP decade the in-window drift is -0.40 bp/meeting with

$t = -0.41$. Fact 2 shows the signed Fed-minus-market gap has a standard deviation of 10 bp. §8.4 shows that neither candidate market-side r^* measure is usable. Taken together:

- Part 1 is estimated on the decade in which the yield response is indistinguishable from zero, so its power is low by construction and it is informative mainly as a null.
- The dispersion half of Part 3 is Cao, Crump, Eusepi & Moench (SR 934) with a different dispersion series; the signed half has no usable variation in the post-2012 sample.
- Part 2 — the maturity signature, together with its nominal/TIPS split — is the one element that has power on assembled data, can fail, and is not occupied.

The prospectus is therefore organised around Part 2 as the load-bearing test of Component A, with Parts 1 and 3 stated as bounded exercises whose most likely outcome is a reportable null, and with the pre-2012 Tealbook sample promoted from a contingency to the condition under which the projection-based programme has power at all (§9). A second, methodologically independent component (§7) requires none of Component A's assumptions.

2 Three motivating facts

All three are established on data already assembled: Gürkaynak–Sack–Wright zero-coupon yields, the Adrian–Crump–Moench daily decomposition, a hand-built FOMC calendar of 300 scheduled meetings and 73 unscheduled actions covering June 1989 to July 2026, the SEP dot-plot archive from January 2012, and the New York Fed Survey of Primary Dealers longer-run federal funds rate from 2013.

2.1 Fact 1: long yields fall only when the Fed is in the room

Between June 1989 and June 2021 the ten-year yield fell 6.99 pp, of which 6.16 pp (88%) accrued inside three-day FOMC windows representing 11.8% of trading days. Outside those windows the drift is -0.012 bp/day with $t = -0.18$ across 7,035 days; a randomisation test placing 324 randomly timed three-day blocks on non-FOMC days puts the observed figure at the 0th percentile of 5,000 draws. The object here is a *conditional mean by day type*, not a statement about the permanence of innovations. Window and non-window days have nearly identical daily volatility (6.72 versus 5.60 bp) and completely different cumulative outcomes. What distinguishes announcement days is that news arriving on them has had a systematically negative sign for three decades, not that it sticks; §7 develops this. Extending to July 2026 — a period no paper in this literature covers — adds a fact that constrains any mechanism. Over 2022–2026 the Fed raised the target 525 bp and the ten-year rose 2.60 pp; inside FOMC windows it *fell* 1.20 pp. The entire tightening in long rates happened on non-FOMC days, and the in-window series continued to decline throughout (Figure 1).

2.2 Fact 2: the two beliefs move together, and asymmetrically

The FOMC publishes its longer-run federal funds projection four times a year; the New York Fed surveys primary dealers for the same object roughly two weeks before every meeting. Netting the 2% inflation objective from both gives two directly comparable measures of r^* — one the Fed's, one the market's — with no term-structure model anywhere in the construction. Three things are true of them (Figure 2). First, **they are never far apart**: across 48 matched meetings

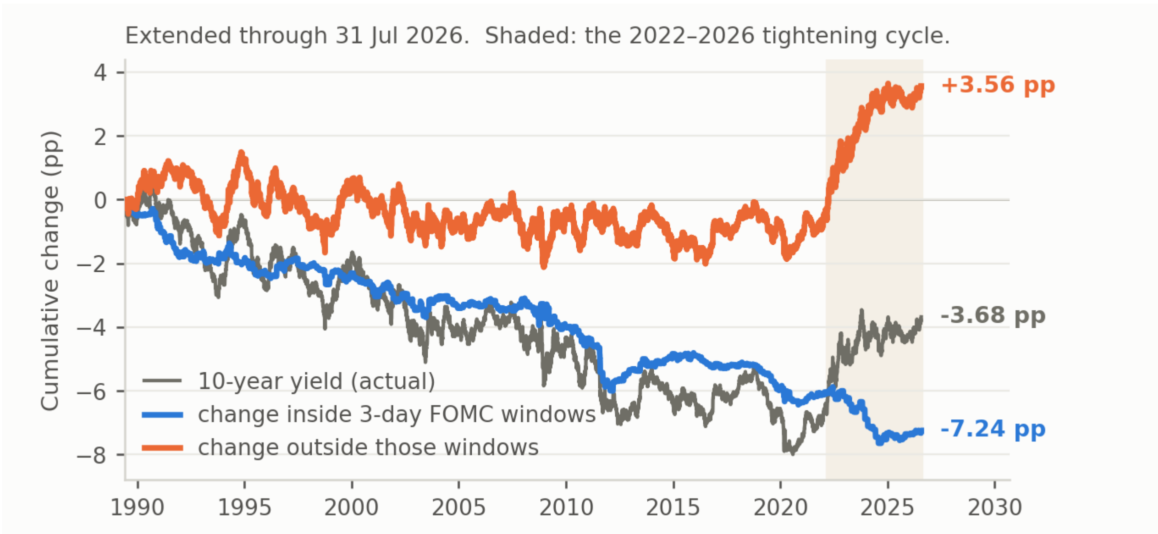

Figure 1: Fact 1. Cumulative change in the ten-year yield from 2 June 1989, split into the part accruing inside three-day FOMC windows and the part accruing on all other days, through 31 July 2026. Shaded: the 2022–2026 tightening cycle.

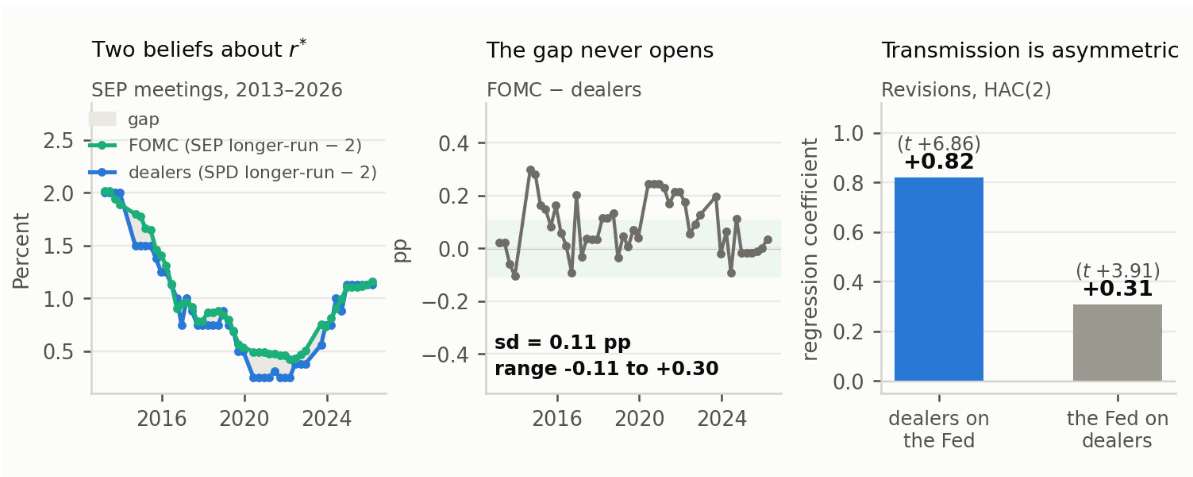

Figure 2: Fact 2. Left: the FOMC's longer-run projection and the primary dealers' longer-run federal funds rate, each less 2.0, at SEP meetings. Centre: the difference. Right: revision regressions, HAC(2), 46–47 meetings.

the Fed-minus-dealer gap has a standard deviation of 0.105 pp and a range of -0.11 to $+0.30$ pp, exceeding 25 bp in absolute value at 4% of meetings. Second, **dealers move with the Fed nearly one-for-one**: regressing the dealer revision on the Fed's revision at the preceding meeting gives $\beta = 0.82$ ($t = 6.86$). Third, **the reverse relation is present but much weaker**: regressing the Fed's revision on the dealer revision already known to it gives $\beta = 0.31$ ($t = 3.91$). The asymmetry is the interesting part, and it is not mechanical: both parties have the opportunity to learn from the other, since the Desk briefs the FOMC with survey results before each meeting. But the evidence stops short of a clean lead-lag. In a horse race the dealer revision loads on both the preceding dot revision (0.60, $t = 3.51$) and the contemporaneous one (0.58, $t = 3.23$), so part of what looks like following is anticipation — unsurprising, since the Fed's own revisions

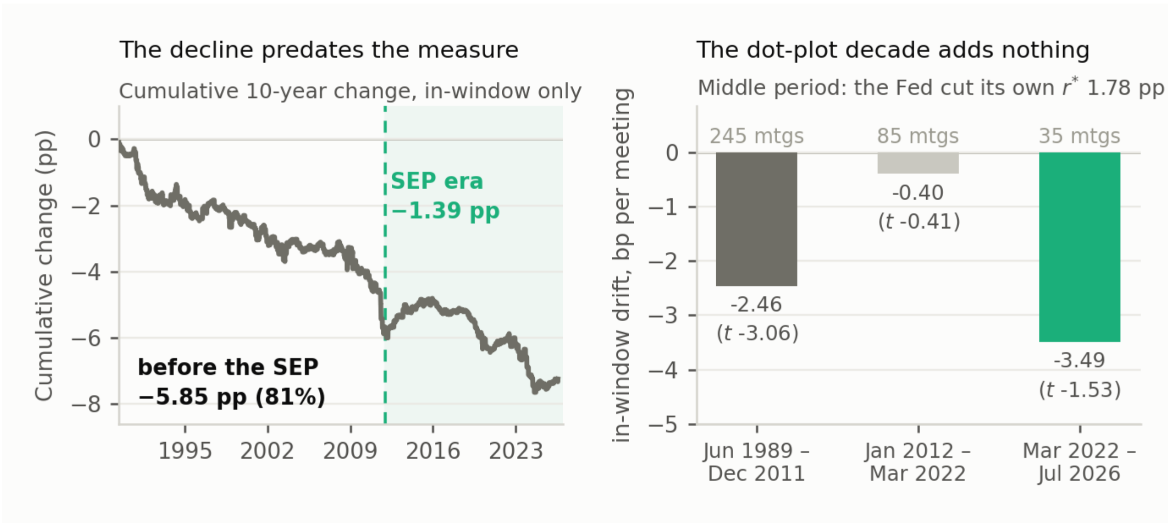

Figure 3: Fact 3. Left: cumulative in-window decline in the ten-year yield with the SEP era shaded. Right: in-window drift per meeting by period.

are serially correlated ($\rho = 0.38$, $t = 2.85$) while dealers' are not ($\rho = -0.07$). Both series are observed eight times a year, which is why the simultaneity cannot be resolved within them; §6.1 states what can and cannot be done about this, and the answer is more limited than an earlier draft of this document claimed.

2.3 Fact 3: the phenomenon predates the measure

Splitting the in-window drift by period gives -2.46 bp/meeting ($t = -3.06$) before 2012, -0.40 ($t = -0.41$) from January 2012 to March 2022, and -3.49 ($t = -1.53$) thereafter. In cumulative terms 81% of the in-window decline is complete before the dot plot begins (Figure 3). In the decade when the FOMC cut its own longer-run rate by 1.78 pp — when an information channel should be loudest — the in-window contribution is indistinguishable from zero. This is a binding constraint on the project rather than a puzzle to be explained away. Every design that uses the published projection is estimated on a sample in which the phenomenon motivating the project is absent. That is the argument for the Tealbook extension in §9, and it is an argument about the whole of Component A, not about A4 alone.

3 Framework

3.1 What a window response can contain

Define the neutral nominal rate $i_t^* \equiv r_t^* + \pi_t^*$ and the policy gap $\text{gap}_t \equiv i_t - i_t^*$, the stance of policy; under (1), $\text{gap}_t = \phi_\pi(\pi_t - \pi_t^*) + \phi_x x_t + v_t$. Starting from the definition of a long yield as an average of expected short rates plus a residual, and assuming the endpoints are locally martingales,

$$y_t^{(n)} \;=\; r_t^* + \pi_t^* + \frac{1}{n}\sum_{j=0}^{n-1} E_t\big[\text{gap}_{t+j}\big] + TP_t^{(n)}. \tag{2}$$

If the gap mean-reverts as an AR(1) with monthly persistence ρ, the expected-gap term is $\omega(\rho, n)\,\mathrm{gap}_t$ with $\omega(\rho, n) = (1 - \rho^n)/[n(1 - \rho)]$, and the analogous weight for a forward covering periods n_1 through $n_1 + n_2 - 1$ is $\omega^f = \rho^{n_1}(1 - \rho^{n_2})/[n_2(1 - \rho)]$. Differencing over an announcement window,

$$\Delta y_t^{(n)} = \underbrace{\Delta r_t^* + \Delta \pi_t^*}_{\text{endpoint news}} + \underbrace{\omega\, \Delta \mathrm{gap}_t + \mathrm{gap}_t\, \Delta\omega}_{\text{stance and persistence news}} + \Delta TP_t^{(n)}. \tag{3}$$

Four terms, one observable. The claim that window moves are endpoint news requires all three of the others to contribute approximately nothing, and each is a separate assumption. The stance exclusion is usually defended by observing that the pattern survives in the 5-year, 5-year-forward rate, where ω^f is small. How small is an empirical question about ρ:

Half-life of the gap	ω, 2-year	ω, 5-year	ω, 10-year	ω^f, 5y5y
1 year	0.56	0.29	0.15	0.01
2 years	0.73	0.48	0.28	0.09
4 years	0.85	0.67	0.48	0.28

Estimating an AR(1) on a crude gap proxy — the one-year yield less HLW r^* plus 2 — gives a half-life of 4.4 years on the full sample and 4.1 years post-1995. At that persistence a 100 bp revision to the expected stance moves the ten-year about 50 bp and the 5y5y forward about 30 bp, so the far-forward evidence does not by itself separate endpoint news from stance news.

The martingale assumption is load-bearing. Equation (2) gives endpoint news a coefficient of one at every maturity only because r^* and π^* are assumed to follow local martingales. If perceived revisions to r^* are partly transitory — which learning models of the Ahonon–Roussellet type explicitly allow — then endpoint news also enters with a weight of the form $\omega(\rho^*, n)$ for some ρ^* close to but below one. The maturity test then discriminates between two decay rates rather than between flat and declining, and its power depends on how far apart ρ^* and ρ are. This is stated as a limitation up front because it determines what a flat estimated profile is evidence *for*: a near-unit-root endpoint, not necessarily a martingale one.

3.2 The maturity signature, and what it can and cannot separate

Equation (3) yields a test that requires no decomposition and no auxiliary data. An endpoint shock enters $y^{(n)}$ with a coefficient of one at every maturity; a stance shock enters with $\omega(\rho, n)$, which is strictly decreasing in n. So the *shape* of the window response across the maturity spectrum discriminates between them, with a slope pinned down by ρ. Because the discrimination comes from a cross-maturity restriction rather than a model of the pricing kernel, it is immune to the critique in §8.2. Three honest qualifications, each of which a referee will raise. *First, this is close to a factor decomposition.* Over 2–10 years, "flat in n" is approximately the level factor and "declining in n" approximately the slope factor, so the raw test asks whether announcement responses load on level or on slope — which is what Gürkaynak, Sack & Swanson (2005) and Swanson (2021) have been doing under the target/path/LSAP labels for twenty years. The contribution is not the shape test as such but the *restriction*: $\omega(\rho, n)$ is a one-parameter family, ρ is estimable outside the event sample from the gap process, and the resulting curve can be rejected. The over-identifying content is the match between the ρ that fits the maturity profile and the ρ

that fits the gap dynamics. *Second, $\Delta TP^{(n)}$ has an unrestricted maturity profile*, so any observed shape can in principle be rationalised as a term-premium response. The test is informative about the expectations block conditional on the premium response being smooth and monotone; it is not a model-free identification of endpoint news. "Immune to §8.2" is a claim about decomposition, not about identification, and should not be written as more than that. *Third, the TIPS split is what makes the test sharp.* Running the same shape test on the real curve removes $\Delta\pi^*$ from (3) entirely, and the nominal-minus-real difference identifies the inflation-expectation component of the window response directly. That comparison, rather than the nominal shape alone, is the element with no close antecedent.

3.3 Two channels from intercept disagreement, kept distinct

An earlier draft ran two different objects together. They are separated here because the positioning argument in §4 turns on the difference. **Channel 1: a disagreement premium, $\sigma^2_{\delta,t}$.** Dispersion in beliefs about the intercept propagates into the variance of the entire expected policy path, undamped by horizon — unlike stance uncertainty, which is damped by ω. Ahonon & Roussellet (2026) estimate that investor uncertainty about r^* is on the order of ± 170 bp, which makes the magnitude plausible. This object is *permanent and horizon-invariant*: the prediction is a term premium sensitivity that rises with maturity and does not decay. This is the channel that differentiates the project from Caballero & Simsek, whose demand disagreement mean-reverts — and it is also the channel Cao et al. substantially occupy. **Channel 2: a level effect of the signed gap, δ_t.** With $\delta \equiv r^{*m} - r^{*F}$, the market evaluates the stance against its own neutral rate while the Fed sets policy against its own, so the market perceives a stance δ away from the Fed's. Positive δ means policy looks too easy: expected inflation overshoot, an expected later correction, compensation in the inflation dimension. Negative δ means policy looks too tight. This object is *transitory*: it is a perceived error that the Fed is expected to correct, so its effect on yields is hump-shaped in maturity, not flat. The two channels are therefore not in tension — a permanent disagreement about a constant can generate a temporary expected correction — but the prospectus must not use "permanent and horizon-invariant" to defend against Caballero & Simsek and "temporary correction" to generate the hump without saying which object each claim is about. Fact 2 constrains Channel 2 severely: the level of δ measured from published projections never gets large. §6.4 addresses what can be done about that, and §9 states plainly what would have to be true for the channel to remain testable.

3.4 A minimal two-belief model

Let the Fed set policy against its own estimate r^{*F} and let the representative investor hold r^{*m}, with $\delta_t \equiv r^{*m}_t - r^{*F}_t$. The investor knows the rule, so his expectation of the *policy rate* uses the Fed's intercept; but he evaluates the *economy* using his own, so he perceives a stance δ_t looser than the Fed does. If he believes policy that is δ too easy raises inflation, and that the Fed will eventually respond, his expected path carries a correction term:

$$E^m_t[i_{t+j}] \;=\; r^{*F}_t + \pi^*_t + \rho^j\,\mathrm{gap}_t \;+\; \lambda_j\,\delta_t, \tag{4}$$

where λ_j is the perceived-correction profile. Averaging (4) and adding a premium gives

$$y^{(n)}_t \;=\; \underbrace{r^{*F}_t + \pi^*_t}_{\text{endpoint}} + \underbrace{\omega(\rho,n)\,\mathrm{gap}_t}_{\text{stance}} + \underbrace{\Lambda(n)\,\delta_t}_{\text{perceived error}} + \underbrace{TP^{(n)}(\sigma^2_{\delta,t})}_{\text{disagreement premium}}, \tag{5}$$

with $\Lambda(n) = n^{-1}\sum_{j<n}\lambda_j$.

The central assumption, stated as such. In (5) the long-run anchor of the nominal yield is r^{*F}: the *Fed's* estimate, whether or not it is correct. This is a strong assumption and it is the opposite of the accounting argument that the Fed cannot durably move r^* because it does not set it. Both positions are defensible — if the Fed's rule intercept is what generates realised short rates, then the Fed's estimate is the relevant endpoint for pricing regardless of the true r^* — but the project must adopt one and defend it, and this document adopts the former. An alternative in which the endpoint is $\theta r^{*F} + (1-\theta)r^{*m}$ nests both and makes θ estimable from the maturity profile; that generalisation is the natural robustness exercise.

λ_j **is asserted, not derived.** The shape stated in words — near zero at short horizons, peaking at business-cycle horizons, decaying at the long end — is imposed. As written, the hump prediction cannot fail, because any interior maximum confirms it. Making the prediction testable requires deriving λ_j from a stated Phillips curve and rule (ϕ_π, ϕ_x, the inflation persistence), which pins down both the peak location and the magnitude of $\Lambda(n)$ relative to $\omega(\rho, n)$. That derivation is scheduled work, not completed work, and the maturity table below should be read as a set of conjectured profiles until it is done.

Term	**Loading on n**	**Shape**
Endpoint news Δr^{*F}	1 (or $\omega(\rho^*, n)$, $\rho^* \to 1$)	flat, or near-flat if the endpoint is not a martingale
Stance news Δgap	$\omega(\rho, n)$	strictly decreasing, slope set by ρ
Perceived error δ	$\Lambda(n)$	hump-shaped — *conditional on* a derived λ_j; peak location is the falsifiable content
Disagreement premium σ_δ^2	increasing in n	rising, and not decaying — the signature that separates it from every damped uncertainty measure

The restriction that does come free is the sign split. δ enters expected inflation and the expected real path with opposite signs: if policy is perceived too easy, breakeven inflation should rise while the real forward falls at the correction horizon, and the reverse if policy is perceived too tight. So the *split* of the response between the nominal and TIPS curves identifies the sign channel without any decomposition of the nominal curve itself.

3.5 Is "intercept" load-bearing?

Worth confronting directly, because a referee will. Every loading in the table above follows from "there is a long-run anchor and agents disagree about it"; none of them requires that anchor to be the constant in a *reaction function* rather than the endpoint of the short rate process. And since all 594 archived longer-run PCE submissions are exactly 2.0 with zero within-meeting dispersion, "intercept disagreement" reduces empirically to disagreement about the longer-run nominal dot — the object Hillenbrand already uses. The rule framing earns its place in two narrower ways. First, it is what makes the differentiation from Bauer, Pflueger & Sunderam precise rather than rhetorical: same rule, different coefficient. Second, and more aggressively, if the intercept moves and is absorbed into forecaster fixed effects, then their slope estimates are identified only up to that absorption — an intercept that is time-varying and forecaster-specific contaminates $\hat{\phi}_\pi$. That is a stronger claim than "they left the intercept open" and is worth checking against their appendix before it goes in a draft.

4 Relation to the literature

Paper	Object	What it leaves open
Hillenbrand (2025, *RFS*)	Window decline is endpoint learning	Never tests the exclusions; no belief data
Gürkaynak, Sack & Swanson (2005); Swanson (2021, *JME*)	Target/path/LSAP factors from the cross-maturity response to announcements	The factors are statistical; no structural restriction tying the path loading to an estimated ρ; no endpoint object
Nakamura & Steinsson (2018, *QJE*)	Fed information effect: announcements move private forecasts of fundamentals	Horizon is business-cycle, not the endpoint; no long-run projection data
Bauer, Pflueger & Sunderam (2024, *QJE*)	Perceived *slope* ϕ_π from forecaster panels; prices term premia	Absorbs the intercept into forecaster fixed effects, explicitly
Bauer & Rudebusch (2020, *AER*)	r^* *level* in an arbitrage-free model	A single trend, not two contested beliefs
Cao, Crump, Eusepi & Moench (SR 934)	Forecaster-vs-forecaster disagreement about the long-run level; $\sim$1/3 of TP variation	Unsigned; not Fed-vs-market; requires an affine model. *Occupies the Channel-1 result*
Caballero & Simsek (2022, *AER*)	Fed-market disagreement $\rightarrow$ sign-dependent policy risk premium	Disagreement is about *demand*, which mean-reverts; no term structure, no TIPS
Cieslak & McMahon; Cieslak, McMahon & Pang	Perceived policy errors raise term premia; WSJ-text measure	Error is about the state and the slope, never the intercept
Bauer, Lakdawala & Mueller (2022, *EJ*)	Option-implied policy uncertainty, 1–2 year horizon	Silent on r^*; a control, not a competitor
Ahonon & Roussellet (SR 1187)	Investor uncertainty about r^* priced in bonds	Representative investor; no Fed-vs-market object
Savor & Wilson (2013, 2014); Ai & Bansal (2018, *Ecta*); Lucca & Moench (2015, *JF*)	Announcement-day risk premia; bonds earn most of their excess return on announcement days	A *return* premium requiring no offset; silent on the sign of the yield-level change. Directly relevant to §7

Three elements appear to be unclaimed: disagreement that is *Fed-versus-market* rather than forecaster-versus-forecaster; a *signed* gap rather than a dispersion measure; and the mapping from the sign to *which* risk is compensated. The SEP *longer-run* dot as an asset-price regressor is also effectively unused — work using the SEP uses near-term dots or lift-off timing. Three positioning risks require explicit treatment. *Caballero & Simsek* already have a sign-dependent policy risk premium from Fed-market disagreement; the defence is that intercept disagreement is permanent and horizon-invariant whereas demand disagreement mean-reverts. Note that this defence applies to Channel 1 only, and Channel 1 is the one Cao et al. occupy — so the differentiated space is narrower than a list of three unclaimed elements suggests, and the project should lead with transmission and the maturity signature rather than with a dispersion result. *Gürkaynak–Sack–Swanson and Swanson* are the reason §3.2 must be presented as a restricted shape test rather than as a novel cross-maturity design. *Nakamura–Steinsson* is the reason the information-effect reading of Hillenbrand cannot be treated as unexamined — what is unexamined is the *endpoint* version of it.

5 Data

- **Yields.** GSW zero-coupon nominal curve, 1–30 years, daily 1961–2026; GSW TIPS curve from 1999 (usable from roughly 2003); implied breakevens.

- **Decompositions.** ACM daily risk-neutral and term-premium series, used only for comparison with the literature, for the reasons in §8.2; Kim–Wright as the second model required by §8.2.
- **Events.** 300 scheduled FOMC meetings and 73 conference calls and unscheduled actions, June 1989 – July 2026, assembled from Federal Reserve historical meeting pages, with SEP-release meetings flagged.
- **The Fed's belief.** SEP longer-run federal funds projections, 57 meetings from January 2012, at participant level: mean, median, cross-participant standard deviation and interquartile range.
- **The market's belief.** NY Fed Survey of Primary Dealers longer-run federal funds rate, 92 surveys from 2013, with 25th and 75th percentiles; Survey of Market Participants from 2014; merged into the Survey of Market Expectations from 2025.
- **To be added.** Treasury auction calendar for the issuance control; Blue Chip Financial Forecasts long-range; Tealbook/Greenbook longer-run projections (five-year lag, available to 2019); intraday Treasury futures, for the narrower purpose set out in §6.1.

6 Component A: transmission, pricing, and disagreement

6.1 A1. Does the longer-run projection move prices?

The SEP/non-SEP contrast. The longer-run dot is published at four meetings a year and not at the other four. If long-run guidance operates through the projection, the in-window response of long yields should be larger at SEP meetings, and larger still at SEP meetings where the longer-run dot moved. This is a within-Fed comparison that holds the announcement format fixed, and it is estimable immediately on assembled data:

$$\Delta y_t^{(n)} \;=\; \alpha + \beta_1 \mathrm{SEP}_t + \beta_2 \left(\mathrm{SEP}_t \times |\Delta \mathrm{dot}_t^{LR}|\right) + \gamma' Z_t + \varepsilon_t, \tag{6}$$

with Z_t containing the Kuttner (2001) target surprise, the path factor, and pre-meeting controls in the spirit of Bauer & Swanson (2023). *Power, computed before the regression is run.* With 57 SEP meetings, roughly 28 of which carry a non-zero longer-run revision, and a residual window standard deviation of about 7 bp, the minimum detectable effect on β_2 at 80% power is on the order of tens of basis points per percentage point of revision. The first task in this subsection is to compute that number exactly and report it, so that a null is interpretable as a null rather than as an absence of data. If the MDE exceeds the effect sizes implied by Hillenbrand's own estimates, A1 should be reported as underpowered rather than estimated and discussed. *What intraday data can and cannot do — a correction to v1.* The previous draft claimed that intraday Treasury futures "let the response to the longer-run projection be separated from the response to the near-term path." That is wrong. The SEP is released at 2:00 pm simultaneously with the statement: the longer-run dot, the near-term dots and the statement language all arrive in the same tick. Intraday data separates *statement-plus-SEP* from *press conference*, and nothing finer. Separating the longer-run projection from the near-term path requires cross-meeting variation in which dots moved — that is, the same 57-meeting, ∼28-revision panel, which intraday data does not enlarge. Intraday futures are therefore a refinement of the 2:00 pm window and a way to strip the press conference, not a solution to the simultaneity in Fact 2. The simultaneity in Fact 2 is between the FOMC and dealers at a quarterly frequency and is not resolved by any market-price measurement. *Falsification.* If β_2 is zero and the design is adequately powered, the published endpoint does not move prices and the projection-based transmission story is limited to the pre-2012 implicit channel, testable only with Tealbook data.

6.2 A2. Does the belief revision have the maturity signature of an endpoint shock? (core test)

Estimate the window response at every GSW maturity from 1 to 30 years, on the subsample of meetings where the longer-run projection was revised, and test the cross-maturity restriction in §3.2: a flat (or $\omega(\rho^*, n)$, $\rho^* \to 1$) loading for endpoint news against $\omega(\rho, n)$ for stance news, with ρ first fixed at the value estimated from the gap process and then treated as a free parameter. The over-identifying test is whether the two ρ's agree. Run the same test on the TIPS curve, which removes $\Delta\pi^*$ from (3) entirely, and report the nominal-minus-real difference as the inflation-expectation component of the window response. Report the estimated profile alongside the first two principal components of the yield curve over the same maturity grid, so that the reader can see directly how much of the shape result is level-versus-slope and how much is the restriction.

6.3 A3. Is disagreement about the intercept priced? (bounded; largely occupied)

Cao, Crump, Eusepi & Moench establish that long-run-level disagreement comoves with the term premium. This subsection is therefore not a contribution on its own; it is included because the maturity restriction in Channel 1 is a prediction they do not test and because their measure is forecaster-vs-forecaster while the SEP cross-participant dispersion is internal to the policy-maker. Dispersion varies usefully: the dealer interquartile range of the longer-run rate spans 0 to 0.75 pp, and the FOMC's own cross-participant standard deviation spans 0.22 to 0.46 pp. Two specifications:

- *Level.* Regress long-horizon forward term premia on intercept dispersion, orthogonalised against the Bauer–Lakdawala–Mueller option-implied policy uncertainty measure and against the forecaster dispersion used by Cao et al. The coefficient of interest is what survives the Cao et al. control.
- *State dependence.* Interact pre-meeting dispersion with the announcement surprise, in the Bauer–Pflueger–Sunderam design but with the intercept in place of the slope.

The distinctive prediction — and the only element here that is not already claimed — is that the sensitivity should *not* decay with maturity.

6.4 A4. Is the response asymmetric in the sign of the perceived error? (contingent on Tealbook)

Fact 2 has ruled out the simplest version: with a Fed-minus-dealer gap whose standard deviation is 10 bp, a signed level regressor has no usable variation in the post-2012 sample. Three routes remain, in descending order of confidence:

1. **The pre-2012 sample.** Tealbook longer-run projections against contemporaneous Blue Chip long-range forecasts, where the disinflation and the productivity boom plausibly generated genuine level disagreement, and where the yield response actually lives. This is now the primary route rather than the third, for the reason given in Fact 3.

2. **Distributional gaps.** The Fed publishes the full distribution of participants' longer-run dots and the SPD publishes response percentiles. Tail disagreement may be substantially larger than median disagreement even when the medians coincide. Cheap to check and worth checking before anything else in this subsection.
3. **Revision direction rather than level.** Test whether upward and downward revisions to the longer-run projection have different pass-through to breakevens versus TIPS. Power is low — 18 upward revisions gives 21–31% against plausible effect sizes — so this is a reported null at best.

If routes 1 and 2 both fail, A4 is dropped rather than estimated on 18 events.

7 Component B: why announcement-day news has one sign

Component A rests on a chain of maintained assumptions. Component B rests on almost none.

7.1 Respecifying the question

An earlier draft framed this as "why do only scheduled-date price changes persist?" That is not what Fact 1 establishes, and the distinction matters. Fact 1 compares *conditional means* by day type: -0.644 bp/day in window against -0.011 outside. A driftless random walk has fully permanent innovations and cumulates to zero, so non-window days can have a permanent share of 100% and still contribute nothing to the secular decline. Permanence and drift are different objects, and a variance-ratio decomposition measures the first while Fact 1 is about the second. The puzzle in Fact 1 is **one-sidedness**: why has news arriving on 12% of trading days carried a systematically negative sign for three decades, when those days are no more volatile than any other? Stated that way the question is about expectations formation or about a risk premium earned at announcements, and it connects directly to a literature the previous draft did not engage — Savor & Wilson, Ai & Bansal, and Lucca & Moench, who document that bonds earn a disproportionate share of their excess return on announcement days. Their object is a *return* premium, which requires no offset and implies nothing about the sign of the yield-level change; Hillenbrand's object is a cumulating *yield-level* effect, which is only coherent if the yield is non-stationary over the sample. Reconciling those two is the contribution, and the 2022–2026 extension in Fact 1 — where the outside-window series reverses to $+3.56$ pp — is the first evidence that the offset arrives.

7.2 Designs

Event-time conditional mean path. Estimate the mean daily change in the ten-year yield at each event time from $t-10$ to $t+10$ around scheduled meetings, unconditionally, with block-bootstrapped bands. Nobody has done this for US Treasuries: Pan & Peng report $t-1$, t and $t+1$; Lucca & Moench's $\pm$5-day check is on equities; Hillenbrand aggregates everything outside the three-day window into one bucket. The shape discriminates directly between a build-then-resolve mechanism and the drop-then-rebuild pattern Bauer, Lakdawala & Mueller document for short-rate uncertainty. *Variance ratio by day type.* Still worth running, but as a distinct object with a distinct target: it adjudicates Hanson, Lucca & Wright (2021), who find announcement-window moves largely reverse, against Hillenbrand, who requires that they do not. It does not test

Fact 1. *Stationarity accounting.* Three papers assert cumulating, non-offsetting window effects — two downward, one upward — and none confronts the arithmetic. Report the offset explicitly by sub-period, including post-2021.

7.3 The competing explanation

Lou, Pintér, Üslü & Walker (BIS WP 1232) document supply-driven pre-announcement drift in UK gilts and report that the US effect concentrates in FOMC windows *without* long-dated Treasury issuance. If correct, the mechanism is dealer inventory rather than monetary policy. This test is cheap but it is *not* decisive either way, and the previous draft overclaimed. Treasury issues 10-year and 30-year paper on a fixed monthly refunding cycle, so overlap with the FOMC calendar is mechanical and correlated with the mid-month timing of major data releases. A null is low power; a positive result is confounded with the calendar. The first step is therefore to count how many of the 373 windows contain long-dated issuance and to check whether the split is balanced across the sample and across the release calendar. If it is not, the test is reported as suggestive and the auction-date result is used as a control rather than as an adjudication.

8 Completed work

Four results are finished and require no further data.

8.1 The literature's disagreement about the channel is an event-list artifact

Two circulating papers give opposite answers to whether the window decline is expectations or term premium. I reproduce Pan & Peng's day-$(t-1)$ estimates on their specification to three coefficients (-0.785 / -0.716 / -0.070 against their -0.79 / -0.71 / -0.08) and then show the attribution is not robust: the term-premium share moves from 91% to 46% on the same day, same data and same decomposition, purely by changing the event list. The 73 conference calls and unscheduled actions a scheduled-only filter removes carry -1.78 pp of expectations decline on day $t-1$ alone — -2.5 bp per event against -0.07 bp for scheduled meetings. Neither paper discusses the event list.

8.2 The standard decomposition cannot arbitrate

In a spanned affine model the expectations component is a fixed linear function of the curve, $\Delta rn_t^{(n)} = c_n' \Delta y_t^{\mathcal{N}}$ with $c_n' = b_n' \mathbf{B}^{-1}$; empirically $R^2 = 1.00000$ over 9,282 daily observations. Event responses therefore satisfy $\beta^{rn} = c'\beta^{\mathcal{N}}$ exactly — predicted $+2.5$ and $+29.3$ bp against directly estimated $+2.6$ and $+29.2$ — so the decomposition adds no event-specific information. And $c'\mathbf{1} = 0.529$: a permanent, fully credible parallel shift is reported as 53% expectations and 47% term premium. The algebra is a corollary of Joslin, Singleton & Zhu (2011) and the bias is Bauer & Rudebusch (2020); the contribution is the explicit filter for ACM and the event-study corollary. Two things are required before this is written up. *First*, report $c'\mathbf{1}$ for Kim–Wright as well as ACM; a result about "the standard decomposition" that rests on one model repeats the error the project criticises elsewhere. *Second*, preempt the obvious referee response — that the spanning identity is a property of *spanned* models and weakens under the unspanned specifications of Joslin, Priebsch & Singleton (2014). The honest scope of the claim is: for the spanned models actually used in this literature, including ACM, the decomposition carries no event-specific information beyond the curve.

8.3 The effect and the measurement do not overlap in time

Fact 3 above: 81% of the in-window decline predates the dot plot.

8.4 Neither candidate market-side r^* measure is usable

The ACM 9-year, 1-year-forward risk-neutral rate loads 0.287 on the one-year yield with $R^2 = 0.83$ and a level correlation of +0.984; it is a damped copy of the policy cycle, and under a stationary VAR no forward horizon isolates an endpoint. The survey measure fails for the opposite reason: dealers load 0.82 on the Fed's own revision, so it is contaminated by the signal it is meant to price. One is circular with yields, the other with the Fed. This closes the Fed-market wedge design rather than leaving it open, and it is a reportable negative result.

9 Risks and what would kill each component

- **Power in the SEP era, and the Tealbook dependency.** 57 meetings, of which four a year publish the longer-run dot, in a decade where the in-window yield response is -0.40 bp/meeting with $t = -0.41$. Tealbook is not a contingency for A4; it is the only sample in which *any* projection-based design is estimated on the period containing the phenomenon. If archive access does not materialise, Component A reduces to A2 estimated on a weak sample plus a reported null, and Component B becomes the dissertation.
- **The maturity test may not separate.** If the perceived endpoint is not a martingale, the test discriminates two decay rates rather than flat against declining, and 28 revisions may not distinguish them. Report the implied ρ^* confidence interval; if it contains values consistent with stance news, the test has failed and should be said to have failed.
- **Simultaneity is not resolved by intraday data.** Fact 2 shows transmission runs both ways at a quarterly frequency between two survey objects. No market-price measurement addresses that. The direction of belief transmission may simply not be identified in this sample, in which case Fact 2 is reported as a descriptive asymmetry.
- **The signed channel may be untestable.** If distributional gaps are as small as median gaps and Tealbook access does not materialise, A4 is dropped.
- **Positioning.** If Caballero & Simsek's empirical work already uses the longer-run dot — which I have not been able to verify — the contribution narrows to the maturity signature and the TIPS asymmetry. Separately, Cao et al. occupy Channel 1 and Gürkaynak–Sack–Swanson occupy the unrestricted version of the shape test; both need explicit differentiation in any draft.

10 Timeline

Period	Component	Task
Weeks 1–4	—	Draft the event-list reconciliation note (§8.1–§8.3): self-contained, model-free, submittable now. Add Kim–Wright $c'\mathbf{1}$ and the unspanned caveat first
Weeks 2–4	B	Count long-dated issuance overlap with the 373 windows; run the issuance test only if the split is usable
Weeks 3–8	B	Event-time conditional mean path, $t-10$ to $t+10$, with block-bootstrapped bands — the core Component B object
Weeks 4–6	A	Derive λ_j from a stated Phillips curve and rule; compute the A1 minimum detectable effect and report it
Weeks 6–14	A2	Maturity-signature test on nominal and TIPS curves, with the PC benchmark
Weeks 8–12	A1	SEP/non-SEP contrast, reported as a null if underpowered
Term 2	B	Variance-ratio decomposition by day type; stationarity accounting; Hanson–Lucca–Wright reconciliation
Term 2	A3	Dispersion and term premia, orthogonalised against Cao et al. and BLM
Contingent	A4	Tealbook and Blue Chip, subject to archive access

11 Requests

1. **The reconciliation note.** §8.1 plus §8.2 is finished, model-free, and adjudicates a live disagreement between two circulating papers. My inclination is to send it as a standalone short piece rather than bury it as a methodological section; I would value a view on that, and on whether the Kim–Wright and unspanned caveats are sufficient scope-limitation.

2. **Tealbook and Blue Chip access.** These bind on the whole of Component A, not on A4 alone. Fact 3 says the phenomenon is pre-2012; every projection-based design is otherwise estimated off-sample. This is the single highest-value resource request in the document.

3. **Structure.** My current view is: lead with the note; make Component B the first chapter, respecified around one-sidedness rather than persistence; and fold A2 — the maturity signature plus the TIPS split — into it as the section asking whether the one-sided announcement news is endpoint news. That yields one paper with a fact, a test and a mechanism rather than two half-chapters. A1, A3 and A4 then become a second chapter conditional on Tealbook. I would like a view on whether that concedes too much of Component A, since it is the part with the genuinely open position.

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